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Motion of the proton in an asymmetric double minimum potential

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A method different from the semiclassical WKB method was developed for calculating the energy levels of a proton and its penetration through a barrier under an asymmetric double minimum potential. The potential was written as the sum of a parabolic term with minimum at $X=0$ and a Gaussian with maximum at X_0 . The stationary states were found by the variational method using the eigenfunctions of the harmonic oscillator for the parabolic potential. The time dependence of the system was found by expanding the eigenstate of the proton at $t=0$ in terms of the stationary states. A Gaussian centered at the minimum of the left well and with an exponent consistent with the ground state of the proton in this well represented the proton at $t=0$. The proton is found to oscillate anharmonically between the two wells. The penetration through the barrier is conveniently described in terms of the average frequency of oscillation and in terms of the extreme of the expectation value of the proton's position in the second well. Both the average frequency of oscillation and the extent of penetration decrease as the height of the barrier increases from 3.0 to 5.0 kcal/mole. At 6.2 kcal/mole no penetration was observed.

INTRODUCTION

The knowledge of proton transfer processes has been enlarged in recent years by the contributions of Eigen and his school with new experimental techniques for determining the rate of fast reactions.¹

These techniques have made possible the determination of rate constants for reactions as fast as the neutralization of H_3O^+ and OH^- in liquid water and in ice. For a long time this reaction has been known to be unusual. As early as 1941 a mechanism that may involve tunnelling of the proton through a potential energy barrier was proposed by Bell.²

The knowledge of the rate constants in water and in ice has provided the basis for a more detailed model that may account for the unusual behavior of the system.

Eigen¹ has proposed a mechanism involving the proton transfer from the donor H_3O^+ to the intervening water molecules and from the intervening water molecules to the base: After a hydrogen bonded system is formed, the proton is transferred by the shifting of its equilibrium position from the donor to the acceptor. This mechanism accounts for several unusual features that are characteristic of this reaction, e.g., the fact that in ice it is faster than a diffusion controlled process and that it is faster in ice than in liquid water.³ In the mechanism proposed by Eigen the rate determining step seems to be the shifting of the proton along the bond it forms between two oxygen atoms. Several explanations are possible for the large value of the rate constant:

- (a) The shifting of one proton along the hydrogen bond has a very low energy of activation or none at all.
- (b) Several protons are shifted simultaneously with a very low energy of activation or no barrier at all.
- (c) "Tunnelling" of the proton through the barrier occurs.

Knowing the potential energy profile for the various

transfers may help to decide which of these explanations is better.

Many results of molecular orbital calculations regarding these profiles are available.⁴ The direct transfer between the H_3O^+ and the OH^- occurs without barrier for interoxygen distances of 3.1 Å or less.⁵ The transfer of the H_3O^+ and OH^- through several intervening water molecules $\text{H}_3\text{O}^+ \dots (\text{H}_2\text{O})_n \dots \text{OH}^-$ has been found to occur without barrier for a simultaneous transfer when n equals one or two. An energy of activation of 10.4 kcal/mole has been found in the case of three intervening water molecules. These results indicate that, in the mechanism proposed by Eigen, as long as the H_3O^+ is not more than the equivalent of two water molecules away from the OH^- site, shifting of the protons may occur simultaneously without energy of activation. When the distance between the oxygen with three protons and the OH^- site is more than the equivalent of three intervening water molecules, an energy barrier larger than 10.4 kcal/mole is to be expected for a simultaneous proton transfer, but the stepwise mechanism may be possible.

The energy difference between the systems



has been shown to increase with n ⁵ reflecting the separation of the charged species. It may be expected that for a large value of n this energy difference will become independent of n . In this case, for a large n the energy profile of the shifting of the proton from the H_3O^+ to the H_2O neighbor molecule will be symmetric as we expect the energy will be insensitive to the distance between the H_3O^+ and the OH^- . Additional stepwise shifting will bring the protonated water molecule in the chain closer to the OH^- site, until the symmetry of the profile is destroyed and an asymmetric potential will be found.

The potential energy for the shifting of the proton between two water molecules

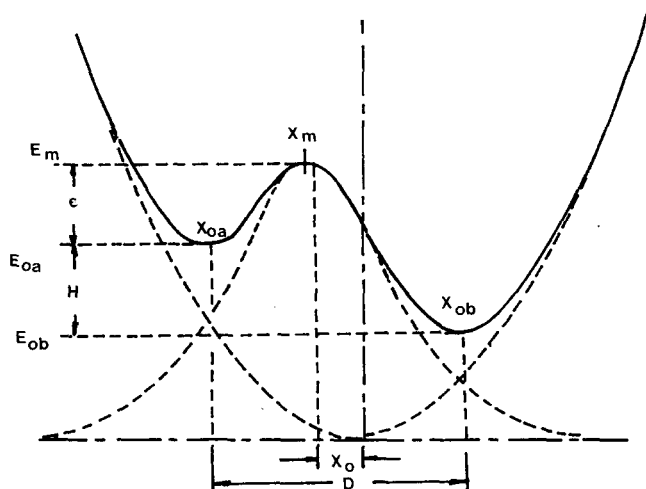
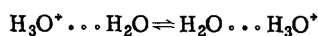


FIG. 1. The solid curve represents the asymmetric double minimum potential as the sum of a parabola and a Gaussian centered at the distance x_0 from the minimum of the parabola. x_{0a} and x_{0b} are the positions of the minima E_{0a} and E_{0b} and x_m is the position of the maximum E_m . D is the distance between the minima.



has been studied in the symmetric case.⁶ At the inter-oxygen distances found in liquid water and in ice the profile has a double minimum with an energy barrier of

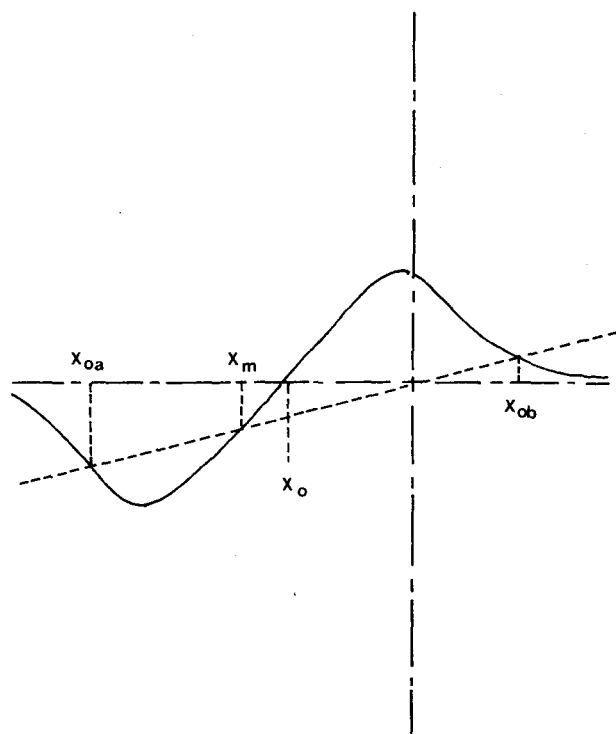


FIG. 2. Determination of the stationary values of the potential function. The broken line is $y = (a_2/\alpha v_0)x$; the solid curve is $y = (x-x_0) \exp(-\alpha(x-x_0)^2)$. The intersections determine the values of x_{0a} , x_{0b} , and x_m from which E^* , H , and D are calculated. For a given value of a_2 there is a unique set of parameters v_0 , α , and x_0 leading to one set of E^* , H , and D .

4.95 kcal/mole. The motion of the proton in this potential has been described before.⁷ The ground state for the protonic vibration was found to be below the barrier, but due to what was described as tunnelling the proton "leaks" through the barrier, oscillating between the two potential wells with a frequency of 10^{13} reciprocal seconds.

When the above shifting occurs closer to the OH^- site, an asymmetric potential with a barrier of 5 kcal/mole or less and an energy change of a few hundredths of an atomic unit are expected. These values will be dependent on how close to the OH^- site the shifting will take place. The motion of the proton under this kind of asymmetric potential has not been studied satisfactorily.

It is the purpose of this report to study it and find whether a nonclassical shifting of the proton through the barrier is to be expected.

CALCULATION OF THE EIGENSTATES

Several methods have been proposed to calculate the eigenstates of an asymmetric double minimum potential. The Wentzel, Kramers, and Brillouin⁸ method does not seem to be appropriate in this case as it applies to situations close to classical mechanics and is valid only for large quantum numbers. Hornig and Sommerjai⁹ proposed the use of a dimensionless Hamiltonian with the potential written as a sum of powers of the distance. At the time their work was published, self-consistent field molecular orbital calculations were not available for systems involving proton shifting between water molecules. We have found it difficult to fit the energy profile of the proton properly with a four degree polynomial. A more convenient fitting could be obtained by the use of a potential formed by the sum of a parabola and a Gaussian. Such potentials have been used before to calculate the eigenstates for the inversion of hydrogen in ammonia¹⁰ and for the proton transfer between two water molecules.⁷ In these cases the potential is symmetric and the maximum of the Gaussian coincides with the minimum of the parabola. If these two points do not coincide, the potential will be represented by an asymmetric double minimum function (Fig. 1):

$$V = a_2 x^2 + v_0 \exp[-\alpha(x-x_0)^2]. \quad (1)$$

The four parameters in the potential function are fitted to the curve obtained from the MO-SCF calculation for several positions of the shifting proton.

The value of a_2 is determined by least squares fitting of the portion of the curve near the oxygen atoms. For this portion the contribution of the Gaussian is negligible. Once a_2 has been found, the other three parameters

TABLE I. Values of the parameters used in the test case.

$a_2 = 0.2755$
$v_0 = 0.5500$
$\alpha = 1.5500$
$x_0 = -0.1200$

TABLE II. Energy levels for the test case. All energy values are referred to the low of the two minima. The last column gives the difference between two successive energy levels within a well.

		E_k	$E_k - E_{k-1}$
1	B_1	0.012883	0.012883*
2	B_2	0.038651	0.025767
3	B_3	0.063893	0.025242
4	B_4	0.088570	0.024707
5	B_5	0.112628	0.024059
6	A_1	0.125130	0.012030*
7	B_6	0.135995	0.023366
8	A_2	0.149190	0.024061
9	B_7	0.158562	0.022567
10	A_3	0.172235	0.023045
11	B_8	0.180166	0.021604
12	A_4	0.193987	0.021752
13	B_9	0.200501	0.020335
14	A_5	0.213747	0.019760
15	B_{10}	0.218874	0.018373
$H = 0.113099$			

*Difference between the first level and the corresponding minimum.

are selected to give the proper values of the energy barrier E^* , the energy difference H between the two minima and the distance D between the minima.

The parameters were calculated by means of an iterative process where E^* , H , and D ; a carefully selected initial set of parameters; and the precision of the desired results are given as input. The positions of the minima and maximum are given by the condition:

$$(a_2/\alpha v_0)x = (x - x_0) \exp[-\alpha(x - x_0)^2]. \quad (2)$$

The stationary values of x are the intersections of the straight line on the left side of Eq. (2) with the function on the right (Fig. 2). With the stationary values of x found this way the calculation of E^* , H , and D is straightforward.

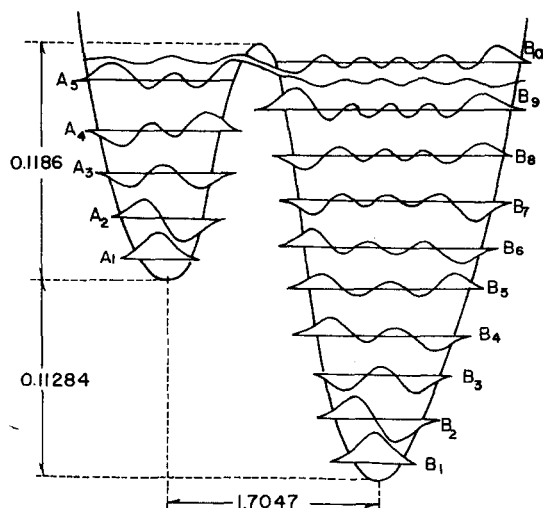


FIG. 3. Potential energy, energy levels, and eigenfunctions for the test case. The energy levels are given in Table II.

In atomic units the Hamiltonian for the system is written as

$$-\frac{1}{2M} \frac{d^2}{dx^2} + a_2 x^2 + v_0 \exp[-\alpha(x - x_0)^2], \quad (3)$$

where the first two terms of the Hamiltonian correspond to a harmonic oscillator of frequency $\nu_0 = \sqrt{2a_2/M}$ in atomic units and centered at $x=0$. The eigenvalues and eigenfunctions of the complete Hamiltonian can be obtained by using the variational method, expanding the eigenfunctions $\Phi_k(\xi)$ in terms of the functions $\psi_n(\xi)$ of the harmonic oscillator of frequency ν_0 . In atomic units the variable ξ is defined as

$$\xi = r^{1/2} x, \quad (4)$$

$$r = \sqrt{2a_2 M}.$$

The matrix elements of the secular equations will be given by

$$H_{m,n} = \nu_0 (n + \frac{1}{2}) \delta_{n,m} + \langle \psi_n(\xi) | v_0 \exp[-\alpha(x - x_0)^2] | \psi_m(\xi) \rangle. \quad (5)$$

In order to facilitate the computation of the matrix elements in a digital computer the following transformations were made:

$$H'_{m,n} = \nu_0 \int_{-\infty}^{+\infty} \psi_n(\xi) \psi_m(\xi) \exp[-\alpha(x - x_0)^2] dx,$$

or

$$H_{m,n} = \frac{\nu_0 N_k N_m}{r^{1/2}} \int_{-\infty}^{+\infty} H_n(\xi) H_m(\xi) \exp\left(-\frac{\alpha}{r}(\xi - \xi_0)^2 - \xi^2\right) d\xi, \quad (6)$$

where $N_k = (r^{1/4}/\pi^{1/4})(k!2^k)^{-1/2}$ is the normalization constant for $\psi_k(\xi)$, $H_k(\xi)$ is the Hermite polynomial of order k and $\xi_0 = r^{1/2} x_0$.

TABLE III. Coefficients of the expansion for the test case of Ψ_0 in terms of Φ_k . The main contribution to Ψ_0 is Φ_6 as expected. There is a small contribution of the excited states in well A. A very small contribution is noted for the states above the barrier. States above the barrier are represented with an asterisk.

C_1	0.000000	B_1
C_2	0.000000	B_2
C_3	0.000000	B_3
C_4	0.000000	B_4
C_5	0.000000	B_5
C_6	-0.999130	A_1
C_7	0.000000	B_6
C_8	-0.039687	A_2
C_9	0.000000	B_7
C_{10}	-0.009229	A_3
C_{11}	0.000003	B_8
C_{12}	-0.008670	A_4
C_{13}	0.000091	B_9
C_{14}	0.002007	A_5
C_{15}	-0.000072	B_{10}
C_{16}	0.000666	*
C_{17}	0.000546	*
C_{18}	0.000411	*
C_{19}	-0.000104	*
C_{20}	-0.000079	*

TABLE IV. Values of the energy barrier E^* , the energy difference H between the minima, parameters for the potential energy curve, values of the coordinates and energies at the two minima and the maximum (E_{0a} , E_{0b} , E_m .)

	I	II	III	IV
ϵ^*	0.0079	0.00499	0.0079	0.0100
H	0.0000	0.06913	0.0589	0.0526
a_2	0.1010	0.2755	0.2755	0.2755
v_0	0.0613	0.5100	0.5100	0.5100
α	3.0670	0.8100	0.8100	0.8100
x_0	0.0000	-0.0896	-0.0761	-0.0678
x_{0a}	-0.450000	-0.653604	-0.668485	-0.675741
x_{0b}	0.450000	0.714168	0.714516	0.714531
x_m	0.000000	-0.287565	-0.238834	-0.210566
E_{0a}	0.053393	0.511850	0.506930	0.503856
E_{0b}	0.053393	0.442723	0.448037	0.451305
E_m	0.061300	0.516847	0.514892	0.513864

If $a_{k,j}$ is the coefficient of the term ξ^j of the Hermite polynomial of order k , the integral (6) can be written as

$$H'_{m,n} = \frac{v_0 N_n N_m}{r^{1/2}} \sum_{k=[1,0],2}^n \sum_{l=[1,0],2}^m a_{n,k} a_{m,l} I(l, k, \alpha, r, \xi_0). \quad (7)$$

The running indices k and l start in 1 or 0 for odd or even Hermite polynomials and run in increments of 2. The integral to be evaluated is

$$I(l, k, \alpha, r, \xi_0) = \int_{-\infty}^{+\infty} \xi^{k+l} \exp\left(-\frac{\alpha}{r}(\xi - \xi_0)^2 - \xi^2\right) d\xi. \quad (8)$$

By means of the following transformations

$$\begin{aligned} \beta &= [(\alpha/r) + 1]^{1/2}, \quad \delta = \frac{\alpha \xi_0}{r}, \quad y = \beta \xi; \\ y_0 &= \frac{\delta}{\beta}, \quad \omega = \left(\frac{\delta}{\beta}\right)^2 - \delta \xi_0, \end{aligned} \quad (9)$$

the above integral reduces to

TABLE V. Energy levels of the four cases considered. The energies are referred to the low of the two minima. The eigenstates are classified as A or B depending on the well they are localized in. When an eigenstate is below the barrier and is delocalized, the coefficient of expansion of the initial state in terms of the stationary states was used as a criterion to classify the state as either A or B. Eigenstates above the barrier are classified as C. Case I shows $g-u$ symmetry and the eigenstates as classified accordingly.

I		II		III		IV	
$A_1 + B_1$	0.004520	B_1	0.008694	B_1	0.008556	B_1	0.008456
$A_1 - B_1$	0.005838	B_2	0.025573	B_2	0.025115	B_2	0.024805
C_{2g}	0.012778	B_3	0.041475	B_3	0.040597	B_3	0.040003
C_{2u}	0.018553	B_4	0.056109	B_4	0.054604	B_4	0.053546
C_{3g}	0.025835	B_5	0.368759	A_1	0.064073	A_1	0.058343
C_{3u}	0.033615	A_1	0.073963	B_5	0.066586	C_6	0.064098
C_{4g}	0.041883	C_7	0.079169	C_7	0.073447	C_7	0.069457
C_{4u}	0.050504	C_8	0.085714	C_8	0.079803	C_8	0.076339
C_{5g}	0.059409	C_9	0.093157	C_9	0.087435	C_9	0.083929
C_{5u}	0.068539	C_{10}	0.101303	C_{10}	0.095631	C_{10}	0.092175

$$I(l, k, \alpha, r, \xi_0) = \frac{e^\omega}{\beta^{k+l+1}} \int_{-\infty}^{+\infty} y^{k+l} e^{-(y-y_0)^2} dy. \quad (10)$$

The integral (10) is well known:

$$\int_{-\infty}^{+\infty} y^{k+l} e^{-(y-y_0)^2} dy = \frac{\sqrt{\pi}}{(2i)^{k+l}} H_{k+l}(iy_0). \quad (11)$$

This expression is real. Substitution of (11) in (10) and in (7) gives the final expression for the matrix elements:

$$H'_{n,m} = \nu_0 (n + \frac{1}{2}) \delta_{n,m} + \frac{v_0 N_n N_m e^\omega}{r^{1/2}} \sum_{k=[0,1],2}^n \sum_{l=[0,1],2}^m a_{k,n} a_{l,m} \frac{\sqrt{\pi}}{2^{k+l}} H_{k+l}^*(y_0), \quad (12)$$

where $H_{k+l}^*(y_0)$ is a Hermite polynomial of order $(k+l)$ with all the coefficients positive.

The above expression is easily programmed for a digital computer such as the IBM 370-135. The program was written in double precision Fortran IV. The matrix was diagonalized by the Jacobi method. After diagonalization, the eigenvalues and eigenvectors were obtained. A study of the convergence revealed that, in order to obtain an accuracy better than 10^{-6} for the lowest 15 eigenvalues, an expansion of at least 38 basis sets must be used. The program contains subroutines to calculate the values of the eigenfunctions Φ_k and the proton probability density at different positions of the proton.

A test case was run for values of the parameters given in Table I. Figure 1 explains the symbols used. This case shows a rather high energy barrier. With a large barrier, the lowest eigenstates will be localized in either well. The reliability of the method will be confirmed if the energy levels in each well correspond to anharmonic oscillator behavior.

The eigenvalues of the lowest 15 states are given in Table II. The eigenfunctions for these states are plotted in Fig. 3. At each point the values of the eigenfunctions are calculated with five significant figures. With this precision, the eigenfunctions decay smoothly beyond the two extreme turning points. It is seen that all but the

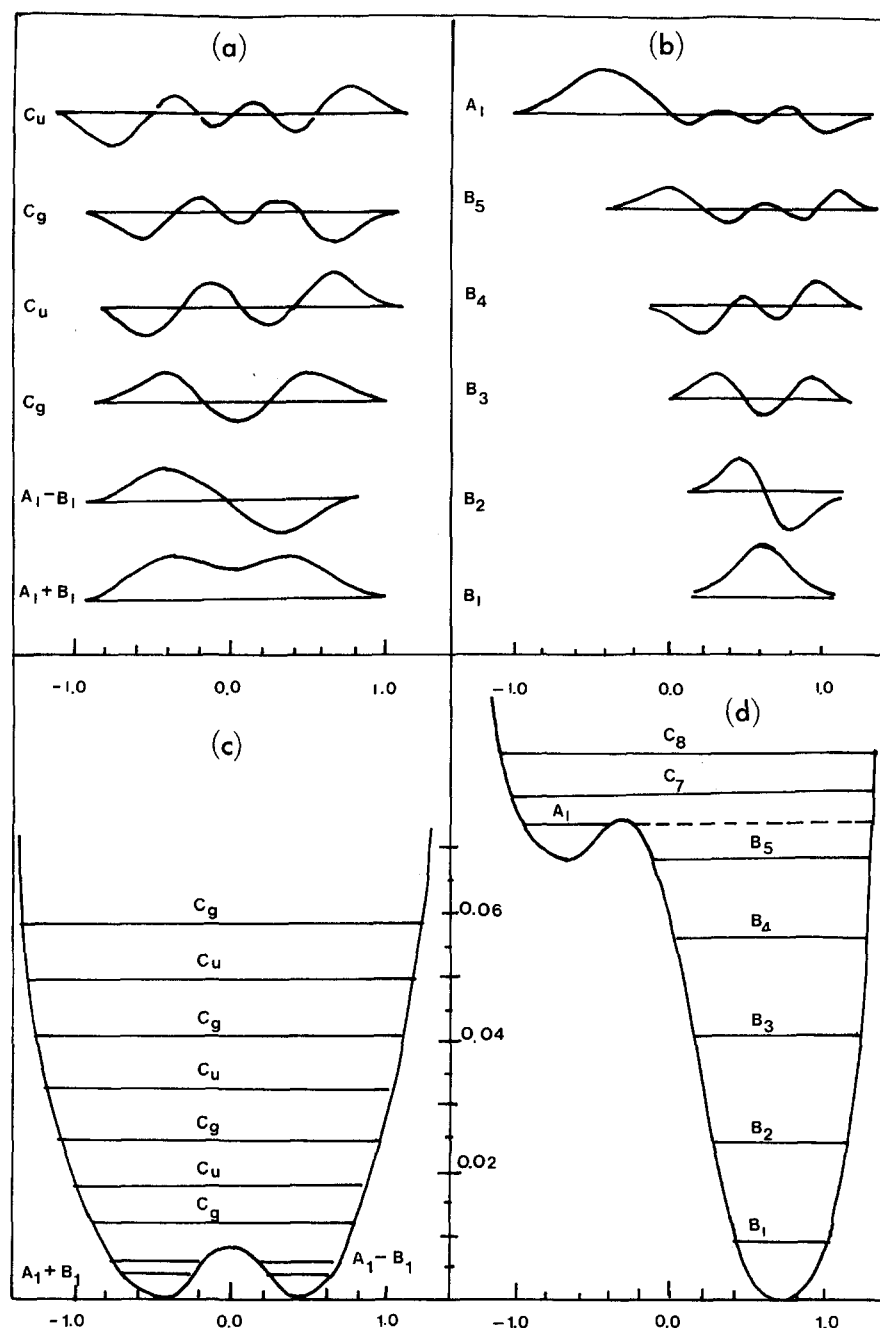


FIG. 4. (a) Eigenfunctions of case I (symmetric, with $E^* = 5.0$ kcal/mole). (b) Eigenfunctions of case II (asymmetric, with $E^* = 3.0$ kcal/mole). (c) Potential energy and energy levels of case I. (d) Potential energy and energy levels of case II.

two eigenstates closest to the top of the barrier are localized in either of the two wells. This allows us to classify each eigenstate as an A or a B eigenstate depending upon whether it is localized in well A or well B . Once the eigenstates were classified as belonging to well A or well B , the differences between the eigenvalues of two successive eigenstates of the same well were calculated. These differences decrease as the energy increases in the way predicted for an anharmonic oscillator.

The plot of the eigenfunctions in Fig. 3 shows that eigenstates B_{10} and A_5 extend in both wells. Each of these two eigenfunctions has 4 nodes in well A and 9 nodes in well B , indicating that they may be approximately written as linear combinations of the fifth and tenth oscillator eigenfunctions centered in wells A and B , respective-

ly. The coefficients for these combinations are not as easily evaluated as in the symmetric case since the gerade-ungerade symmetry has been lost. However, the delocalization of the eigenstates closest to the top of the barrier indicates that the proton has a "nonclassical" behavior as it leaks through the barrier. The description of such leaking is not straightforward as it is in the symmetric case where the "tunnelling" time was calculated as the difference between the gerade and ungerade energies divided by Planck's constant. A different method of presentation of the problem is then required.

TIME DEPENDENCE OF THE SYSTEM

To study the way in which the system changes as a function of time, one may calculate the ground state of

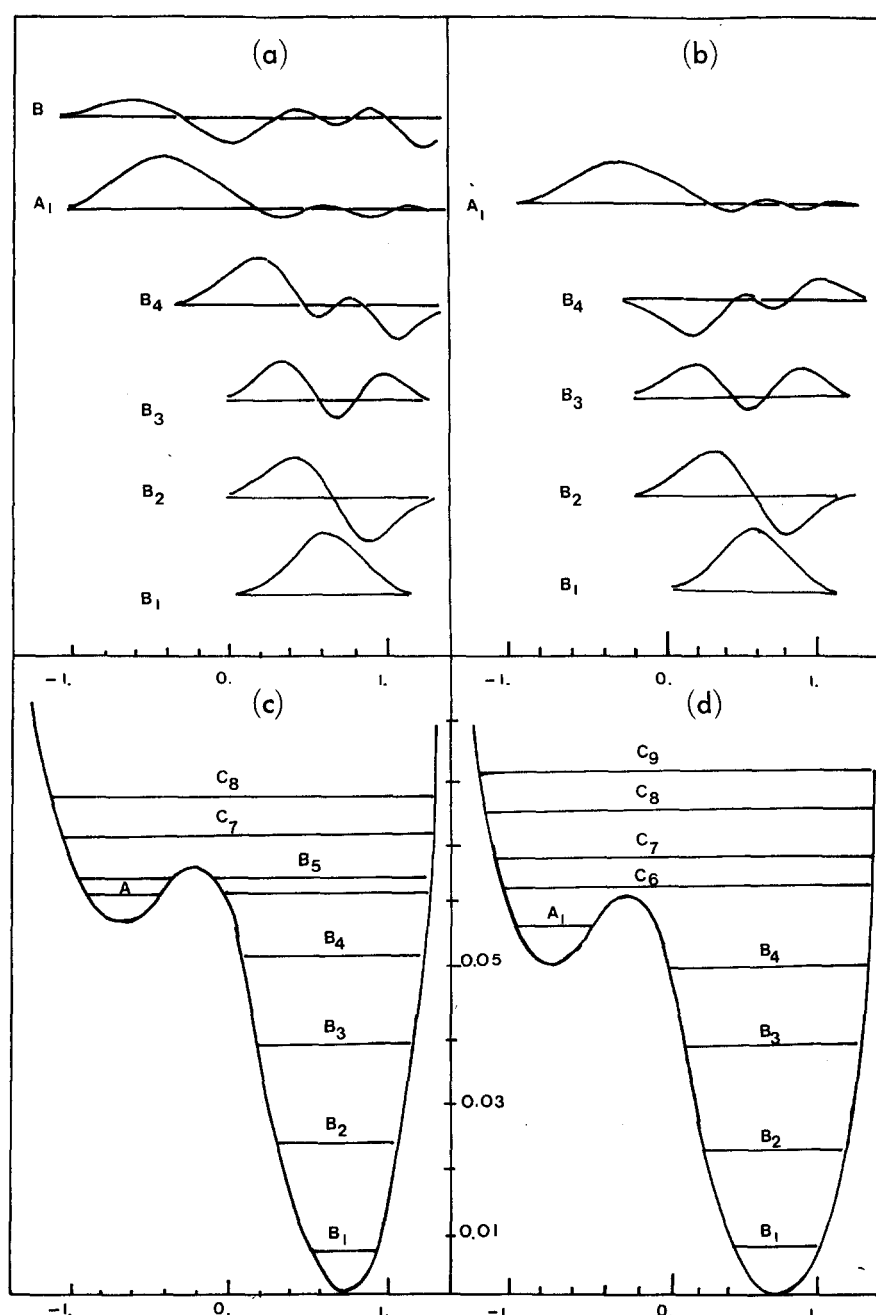


FIG. 5. (a) Eigenfunctions of case III (asymmetric with $E^* = 5.0$ kcal/mole). (b) Eigenfunctions and energy levels of case IV (asymmetric, with $E^* = 6.27$ kcal/mole). (c) Potential energy and energy levels of case III. (d) Potential energy and energy levels of case IV.

the proton in well A, ignoring the anharmonicity contribution and the presence of well B. This can be achieved by calculating with the desired precision the minimum of well A and the second derivative of the potential function evaluated at this point. The derivative will provide the value of k_a , the vibrational constant of the oscillator in well A. The ground state in this case will be given by

$$\Psi_0 = (r_0^{1/4}/\pi^{1/4}) \exp[-\frac{1}{2} r_0(x - x_{0A})^2], \quad (13)$$

where x_{0A} is the minimum of well A and $r_0 = \sqrt{k_a M}$ in atomic units.

This initial state can be expanded in terms of a complete set of stationary states:

$$\Psi_0 = \sum_k c_k \Phi_k \quad (14)$$

where the coefficients $c_k = \langle \Phi_k | \Psi_0 \rangle = \sum_n b_{kn} \langle \psi_n | \Psi_0 \rangle$. b_{kn} are the expansion coefficients of Φ_k in terms of the function ψ_n .

Using the same approach as before, the integral $\langle \psi_n | \Psi_0 \rangle$ can be shown to be

$$\langle \psi_n | \Psi_0 \rangle = \frac{r_0^{1/4}}{r^{1/4} (2^n n!)^{1/2}} \sum_{l=0,1,2}^n \frac{e^\omega}{2^l \beta^{l+1}} a_{nl} H_l^*(y_0). \quad (15)$$

The coefficients of the expansion of Ψ_0 in terms of Φ_k are

$$c_k = \sum_n \frac{b_{kn} r_0^{1/4}}{r^{1/4} (2^n n!)^{1/2}} \sum_l \frac{a_{nl} e^\omega}{2^l \beta^{l+1}} H_l^*(y_0) \quad (16)$$

where all the symbols are defined in the same way as before except for β which is now

TABLE VI. Coefficients of the expansion of $\Psi_0(x, 0)$ as a function of the stationary states ϕ_k . The value of k_{0a} was calculated from the zero point energy of well A.

CASE I			
1	0.675352	13	-0.000180
2	-0.700204	14	0.000085
3	0.223578	15	-0.000064
4	-0.005378	16	-0.000014
5	-0.008671	17	0.000021
6	-0.006359	18	0.000001
7	0.018105	19	-0.000006
8	-0.010745	20	0.000002
9	0.001256	21	0.000000
10	0.001321	22	0.000000
11	-0.000210	23	0.000000
12	-0.000448	24	0.000000
$\sum_k C_k^2 = 0.999999$		$k_{0a} = 0.149729175$	

$$\beta = \left(\frac{r_A}{2r} + \frac{1}{2} \right)^{1/2}. \quad (17)$$

A listing of the lowest 20 coefficients for the test case is given in Table III. The rest of the coefficients are negligible. In this case, the lowest state of well A is practically the only state that contributes to A as the coefficient is 0.9996, smaller contributions from the excited states in well A are due to anharmonic mixing. It should be noticed that the contribution of eigenstates of well B is negligible, indicating that the proton originally placed in well A will remain in this well.

The behavior of the system as a function of time is given by the following expression:

TABLE VII. Coefficients of the expansion of $\Psi_0(x, 0)$ in terms of Φ_k for the asymmetric case with $E^* = 0.00499$.

CASE II			
1	0.000057	20	0.002597
2	0.000420	21	-0.001789
3	0.002477	22	-0.001018
4	-0.014974	23	0.001357
5	0.146294	24	0.000807
6	0.893635	25	0.000627
7	0.362065	26	0.000340
8	-0.176697	27	-0.000282
9	0.075587	28	0.000133
10	0.063183	29	-0.000160
11	-0.060736	30	0.000063
12	-0.047213	31	-0.000126
13	0.031380	32	-0.000039
14	0.020614	33	-0.000121
15	0.014529	34	0.000029
16	-0.010613	35	0.000121
17	0.007620	36	-0.000121
18	-0.005348	37	-0.000104
19	-0.003790	38	-0.000130
$\sum_k C_k^2 = 1.000000$		$k_{0a} = 0.171357918$	

TABLE VIII. Coefficients of the expansion of $\Psi_0(x, 0)$ as a function for the asymmetric case with $E^* = 0.0079$.

CASE III			
1	0.000049	20	0.001635
2	0.000420	21	0.001153
3	0.003063	22	-0.000783
4	-0.028163	23	0.000553
5	-0.877333	24	0.000336
6	0.413642	25	0.000180
7	0.206939	26	-0.000128
8	-0.078633	27	0.000091
9	-0.047592	28	0.000065
10	-0.052759	29	0.000041
11	0.048383	30	-0.000037
12	0.034033	31	0.000026
13	-0.021145	32	0.000026
14	-0.013742	33	0.000019
15	-0.009812	34	-0.000024
16	0.007121	35	-0.000006
17	-0.004988	36	-0.000011
18	0.003412	37	-0.000015
19	0.002349	38	-0.000020
$\sum_k C_k^2 = 1.000000$		$k_{0a} = 0.196623196$	

$$\Psi(x, t) = \sum_k c_k \Phi_k e^{-i(E_k/\hbar)t} \quad (18)$$

where the c_k are the coefficients found earlier. Multiplying the above expression by the complex conjugate one obtains the probability density as a function of time. This probability is shown to be:

$$|\Psi(x, t)|^2 = \sum_k c_k^2 \Phi_k^2 + 2 \sum_k \sum_{l < k} c_k^* c_l \Phi_k^* \Phi_l \cos \omega_{k,l} t \quad (19)$$

where $\omega_{k,l} = (E_k - E_l)/\hbar$.

In order to visualize better the evolution of the system

TABLE IX. Coefficients of the expansion of $\Psi_0(x, 0)$ in terms of Φ_k for the asymmetric case with $E^* = 0.0100$.

CASE IV			
1	0.000034	20	0.000706
2	-0.000356	21	0.000484
3	0.003389	22	0.000303
4	0.059260	23	0.000202
5	0.960856	24	-0.000097
6	-0.213447	25	-0.000098
7	0.149318	26	-0.000046
8	-0.038211	27	-0.000036
9	-0.026017	28	0.000016
10	0.035711	29	-0.000016
11	-0.034110	30	0.000006
12	-0.022960	31	-0.000012
13	0.012860	32	0.000028
14	0.007545	33	-0.000011
15	0.002580	34	0.000002
16	0.003790	35	0.000010
17	0.002580	36	-0.000003
18	0.001656	37	-0.000009
19	-0.001071	38	-0.000019
$\sum_k C_k^2 = 0.999990$		$k_{0a} = 0.246673839$	

TABLE X. Comparison of the average tunneling time obtained by WKB semiclassical approach and the present method.

	I	II	III	IV
Distance between the two inner turning points (A. U.)	0.3942	0.0569	0.2233	0.2680
Average frequency ($\times 10^{-13}$ sec $^{-1}$)	0.91 ^a	3.48	1.67	3.26
Average distance reached beyond the second turning point (A. U.)	0.221	0.198	0.075	...
Average "tunneling" time ($\times 10^{14}$ sec) this calculation	10.99	2.87	5.98	...
Average "tunneling" time ($\times 10^{14}$ sec) WKB semiclassical method	29.64	0.187	5.435	9.35

^aCalculated from $(E_u - E_g)/\hbar$; it gives 0.87×10^{13} sec $^{-1}$.

as a function of time one may calculate the expectation value of the proton's position which will be given by:

$$\langle \Psi(x, t) | x | \Psi(x, t) \rangle = \sum_k c_k^2 \langle \Phi_k | x | \Phi_k \rangle + 2 \sum_k \sum_{l < k} c_k^* c_l \langle \Phi_k | x | \Phi_l \rangle \cos \omega_{kl} t. \quad (20)$$

This value is the sum of two terms, one of which is independent of t while the other is a superposition of waves with amplitudes which are the product of the coefficients of two states and the corresponding transition moment, and with frequencies given by the difference in energy between the two states divided by $\hbar$. In the test discussed, it is clear that the only nonnegligible contribution to the time dependent part will come from the ground state and from the first excited state of well A, indicating a very small variation of the average position of the proton about the equilibrium position of well A owing to anharmonic mixing. It should be noted however that, because all the coefficients are negligible, except the one corresponding to A_1 , this amplitude is very small, indicating that the proton will always remain in well A.

RESULTS AND DISCUSSION

As explained before, a typical energy profile for the shifting of a proton between two water molecules may show an energy barrier $E^* = 5$ kcal/mole or less. The energy difference between the two minima, H , is equal to zero when the transfer occurs far from the OH $^-$ site and is expected to increase as the number of intervening water molecules decreases.

Four cases were selected. In two of them (Cases I and III) the energy barrier was set at 5 kcal/mole with an energy difference between the minima equal to zero in the symmetric case (I) and 37 kcal/mole in the asymmetric case (III). This choice allows us to compare the effect of the symmetry of the potential on the motion of the proton when the energy barrier is the same. Two additional cases (II and IV) with energy barriers of 3.0 and 6.27 kcal/mole also were included. The barriers were obtained by adjusting the parameter x_0 . Increasing the absolute value of x_0 from zero reduces the energy barrier and increases the asymmetry of the system (in-

creases the value of H). This effect is expected to reflect the change in the potential energy profile as the transfer occurs closer to the OH $^-$ site. Once the set of parameters a_2 , α , v_0 , and x_0 was established, the positions of the two minima and of the maximum and the energies at these three points were determined. The eigenvalues for the four cases, obtained by the variational method, were then referred to the value of the energy at the lower minimum. The values of the parameters, the position and energy for the two minima and the maximum for each of the four cases are given in Table IV.

A 38 basis set expansion was used to calculate the eigenvalues and eigenvectors for each of the four cases. This size of basis set gives eigenvalues with accuracy of 10^{-8} or better for the lowest 10 eigenstates.

The eigenvalues for the four cases are given in Table V. Figures 4(c), 4(d), 5(c), and 5(d) display the energy levels and the potential energy curves of Cases I-IV.

The values of the eigenfunctions for different positions of the proton were calculated from the set of eigenvectors. Plots of these eigenfunctions are shown in Figs. 4(a), 4(b), 5(a), and 5(b).

The symmetric Case I was reported in a previous paper⁷ using an 11 basis set expansion. The values of the energies for the four lowest eigenstates, obtained in the present work, agree up to six decimal places with those reported earlier. Both numbers of basis sets give the same splitting between the two energy levels that are below the barrier (0.00132 a. u.), predicting the same frequency (0.87×10^{13} sec $^{-1}$) for the oscillation of the proton between the two wells.

The eigenfunctions reported in the present paper are highly accurate and differ slightly from those obtained with an 11 basis set expansion. The ground state predicted with a 38 basis set expansion shows a maximum at the position of the minimum of each of the two wells. Protonic density in the classically forbidden region is considerable; however, in this section the value of the eigenfunction is a minimum [Fig. 4(a)].

In the first of the asymmetric cases (II) the first five energy levels are localized in well B, only one energy level below the barrier is found in well A, and it extends over both wells. In Case III, which corresponds to a barrier of 5 kcal/mole, the first four energy levels are localized in B. There are two energy levels below the barrier and they extend over both wells. In Case IV (6.27 kcal/mole) only one energy level below the barrier is found in well A. This eigenstate A_1 is mainly localized in well A.

CHOICE OF INITIAL STATE

In the test case discussed before the initial state was chosen by evaluating the second derivative of the potential energy: at the minimum of well A. This choice seems reasonable for energy barriers of 10 kcal/mole or more. For smaller barriers the potential in the vicinity of the minimum of well A differs significantly from that of the harmonic oscillator. This initial state, calculated by means of the second derivative at the mini-

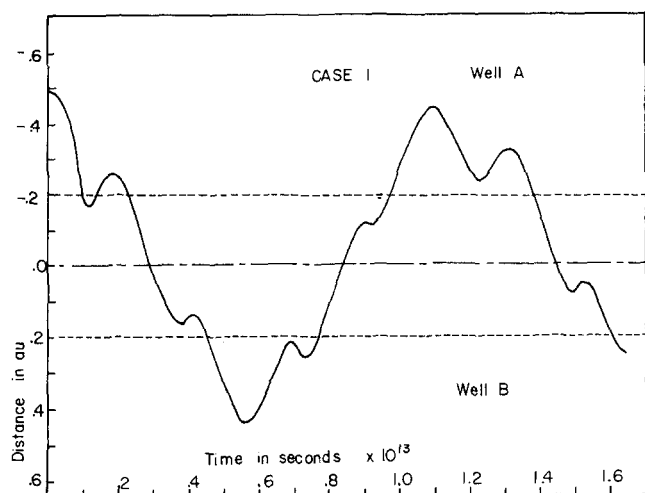


FIG. 6. Expectation value for the proton's position, case I (symmetric with $E^* = 5.0$ kcal/mole).

mum gives in our four cases energies that are above the lowest stationary state in well A (state A_1). In Case II this choice produces an initial energy above the barrier.

A more suitable description for the initial state was obtained by means of a normalized Gaussian centered in x_{0A} and with an exponent consistent with the energy of the eigenstate A_1 . The departure of the eigenfunction Φ_{A_1} from this Gaussian is due to the delocalization of this eigenstate over the two wells. The coefficient of the expansion of the initial state in terms of the Φ_k are given in Tables VI-IX for Cases I-IV.

The table includes the value of k_{0A} used to calculate the exponent of the normalized Gaussian and a check of the normalization of Ψ_0 .

It should be noted that for the asymmetric cases the eigenstate A_1 has the largest contribution to the initial state. However the data show that this initial state is extensively mixed with a large number of eigenstates

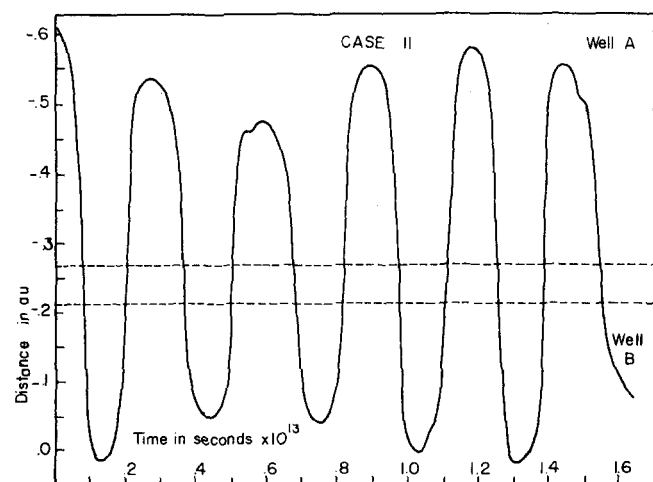


FIG. 7. Expectation value for the proton's position, case II (asymmetric, with $E^* = 3.0$ kcal/mole).

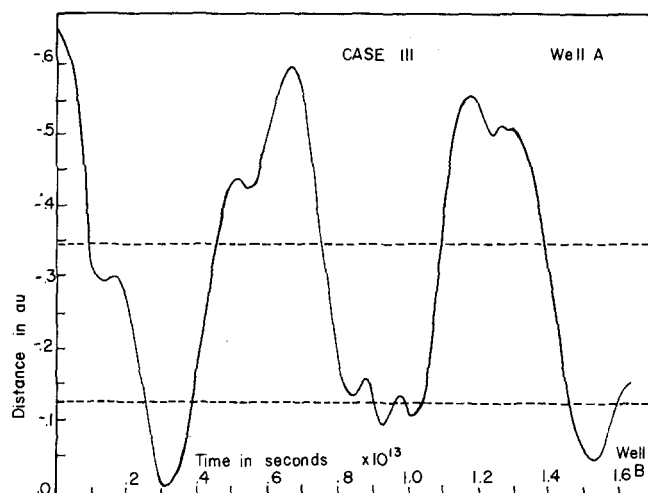


FIG. 8. Expectation value for the proton's position, case III (asymmetric, with $E^* = 5.0$ kcal/mole).

above the barrier. This extensive mixing, found in both the asymmetric and the symmetric cases, prevents us from giving a simple description of the motion of the proton in terms of the contributions of two or three of the stationary states. A complete application of Eq. (20) is necessary in order to calculate the change of the expectation value of the proton's position as a function of time.

The results of the motion of the proton for the symmetric case are shown in Fig. 6. Because of the mixing with the upper eigenstates, the motion is not simple harmonic as implied in earlier publications where the frequency of oscillation between the two wells was found as the difference between the energies of the doublet divided by Planck's constant. However, the motion possesses the symmetry expected in this case. It should be noted that the extreme position reached by the proton in either well is slightly less than the position of the minimum and that the expectation value of the proton's position will be

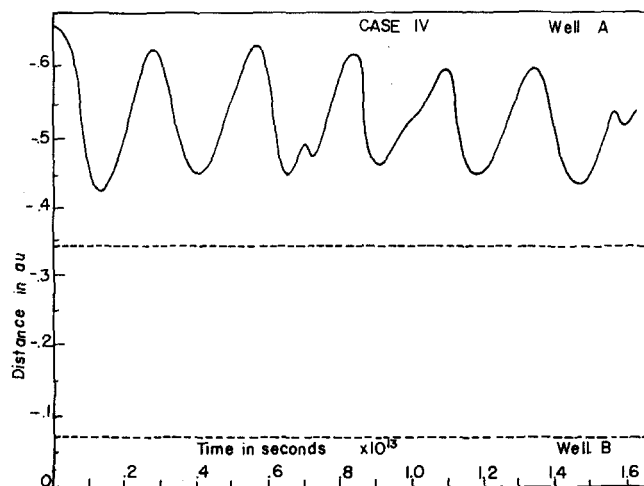


FIG. 9. Expectation value for the proton's position case IV (asymmetric, with $E^* = 6.27$ kcal/mole).

within the classically forbidden region approximately $\frac{1}{3}$ of the time and $\frac{1}{3}$ in each of the wells. This result is consistent with the eigenfunction found for the ground state in the symmetric case.

Because of the anharmonicity of the motion it is not possible to give a frequency; however, we have found it convenient to define the average frequency as the number of complete oscillations per unit time taken over an extended period of time. A similar approach was taken in order to calculate the average distance the proton reaches beyond the turning point of well *B*. In this case the average frequency is found to be $0.91 \times 10^{13} \text{ sec}^{-1}$ (Table X) as compared to $0.87 \times 10^{13} \text{ sec}^{-1}$ for the frequency calculated from the splitting of the *g*-*u* doublet. This result gives an average "leaking time" of $10.99 \times 10^{-14} \text{ sec}$.

The motions for the asymmetric cases are shown in Figs. 7, 8, and 9. For a barrier of 3 kcal/mole the ground state of well *A* is very close to the top of the barrier (0.102 kcal/mole) and the distance between the two inner turning points is only 0.0569 a.u. The proton oscillates between the two wells with an average frequency of $3.48 \times 10^{13} \text{ sec}^{-1}$; the average distance the proton reaches beyond the turning point of well *B* is 0.198 a.u. and the average leaking time is $2.87 \times 10^{-14} \text{ sec}$. For a barrier of 5 kcal/mole (Fig. 8) the ground state of well *A* is 1.743 kcal/mole below the barrier and the distance between the turning points is 0.2233 a.u. The motion is highly anharmonic. The average frequency for the oscillations of the proton between the two wells is $1.67 \times 10^{13} \text{ sec}^{-1}$; however, in each oscillation the proton reaches on the average only 0.075 a.u. beyond the second inner turning point. The average leaking time for this case is $5.98 \times 10^{-14} \text{ sec}$, about twice the value of the preceding case.

These results show that in order to evaluate the leaking of the proton through a barrier two factors should be considered: the frequency of the oscillation and how far the proton penetrates within the second well. The factor is critical in Case IV (Fig. 9) with 6.27 kcal/mole,

where the average frequency of oscillation is $3.26 \times 10^{13} \text{ sec}^{-1}$ but the expectation value of the proton's position never reaches the first turning point of the barrier. It remains confined to the first well where it oscillates due to anharmonicity.

For comparison the "tunnelling times" were calculated using the semiclassical WKB Method. The results are shown in Table X, which shows no correlation between the semiclassical WKB results and the present work. Very significantly WKB Method predicts a tunnelling time of $9.35 \times 10^{-14} \text{ sec}$ for a barrier of 6.25 kcal/mole. While the present method demonstrates that the average position of the proton does not penetrate the barrier at all.

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